

Test Plan

Project: Electronics Store Inventory Management System

Phase 1 — Test Plan

Item	Details
Version	1.0
Team	H9
Technology Stack	Python (Django)
Status	Draft / For Review

Team Members

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Part of the Team H9 Phase 1 document set: 01 SRS | 02 Test Plan | 03 Test Cases | 04 Architecture & Design.

Section and requirement numbers follow the master SRS so cross-references (RTM, VFR, DS-xx, TC-xx) remain valid across the four documents.

5.1 Security

5.1.3 Penetration Testing Plan

Penetration testing is planned for the implementation and testing phase only. No penetration-testing results are claimed in Phase 1. Testing will be performed only against the authorized project development/test environment.

The planned penetration-testing scope covers authentication bypass, authorization and privilege escalation, session management, CSRF protection, malicious or invalid input handling, SQL-injection-style input testing, password/security-secret exposure, and access control for inventory-changing operations.

Planned Security Test	Mapped Requirement	Expected Validation
Authentication bypass	ESIM-SR-001	Protected pages cannot be accessed without authentication.
Authorization / privilege escalation	ESIM-SR-002	A user without the required role cannot perform restricted inventory operations.
Password / secret exposure	ESIM-SR-003, ESIM-SR-006	Passwords and authentication secrets are not stored or logged in plaintext.
CSRF testing	ESIM-SR-004	Unauthorized state-changing requests are rejected.
Malicious / invalid input testing	ESIM-SR-005	Invalid or malicious input is rejected without corrupting valid records.
Session-management testing	ESIM-F-002	After logout, protected functionality requires re-authentication.

6. Quality Attributes & Acceptance Tests

Acceptance is based on measurable criteria rather than only visual inspection. Phase 1 defines the planned tests; actual pass/fail evidence will be produced during implementation and testing.

6.1 Functional Acceptance Suite

- Authentication and logout.
- Add, view, search/filter, update and delete component records.
- Availability and low-stock calculation at boundary quantities.
- Dashboard totals and indicators.
- Inventory report generation.
- Input validation and error handling.
- Authorization of restricted operations.
- Persistence of valid inventory records.

6.2 Performance Acceptance Suite

- Measure response time for inventory list, search, add, update and delete operations under defined project load.
- Verify that at least 90% of selected normal-load requests meet the 2-second target.
- Verify that the system can handle at least 1000 component records in the test database.

6.3 Security Acceptance Suite

- Attempt protected-page access without login.
- Attempt restricted inventory operations using an unauthorized role.
- Inspect password storage and confirm plaintext passwords are not stored/logged.
- Test CSRF protection and server-side validation.
- Check logs for absence of authentication secrets.

6.4 Usability & Compatibility Acceptance Suite

- Complete common inventory tasks using the primary UI without developer instructions.
- Verify clear success and error feedback.
- Run core workflows on current Chrome, Edge and Firefox versions.

6.5 Exit Criteria

- All High-priority functional requirements have planned validation coverage.
- No critical requirement is left without a corresponding test/validation criterion.
- All required security controls have been tested before final acceptance.
- The RTM shows traceability for all planned requirements.
- Any deferred or out-of-scope functionality is documented.